

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: Batch 13

Start Date: 2 February 2026

Names: BHARATAKUMAR G NADAKUR

Internship Duration: 8 weeks

1. Project Title

StockTrendAI — Stock Trend Prediction Dashboard for INFY (NSE)

2. Project Objective

To build a machine learning model that predicts whether INFY stock will go UP or DOWN the next day, and deploy it as an interactive web dashboard using Streamlit. The project uses historical stock data (2017-2022) and technical indicators like RSI, MACD, and Moving Averages to train a Random Forest Classifier.

3. Project description in detail

We built a stock trend prediction system for Infosys (INFY) listed on NSE. The dataset had 1,236 rows of daily OHLCV data from Yahoo Finance (2017-2022). We added technical indicators (MA10, MA20, RSI, MACD, Volatility) as features and created a binary target: 1 if next day close is higher, 0 otherwise. A Random Forest Classifier was trained on 80% of the data and tested on 20%, achieving ~55% accuracy. The final output is a Streamlit dashboard with 4 tabs: Data & Charts, Technical Indicators, Model Results, and a live Prediction tool. Tools used: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, and Streamlit.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand problem statement, explore dataset, set up Python environment	Downloaded INFY.NS.csv from Yahoo Finance, explored data columns and shape
Week 2	Data cleaning: check null values, handle missing data, sort by date	Confirmed no null values, sorted data by date, saved cleaned CSV
Week 3	Feature engineering: add MA10, MA20. Exploratory Data Analysis	Computed MA10 and MA20, plotted closing price and volume charts
Week 4	Handle outliers using IQR, prepare train-test split	Capped outliers in Volume, split data 80/20 for training and testing

Week 5	Train baseline Logistic Regression model, evaluate accuracy	Trained Logistic Regression, got 51% accuracy - identified class imbalance issue
Week 6	Add RSI, MACD, Volatility features. Switch to Random Forest	Added 5 new features, trained Random Forest, improved to ~55% accuracy
Week 7	Build Streamlit dashboard with charts and technical indicator tabs	Completed 4-tab dashboard: Data, Indicators, Model Results, Predict
Week 8	Add prediction form, confusion matrix, feature importance chart	Prediction tab working with confidence score, final model integrated

5a. Key Milestones

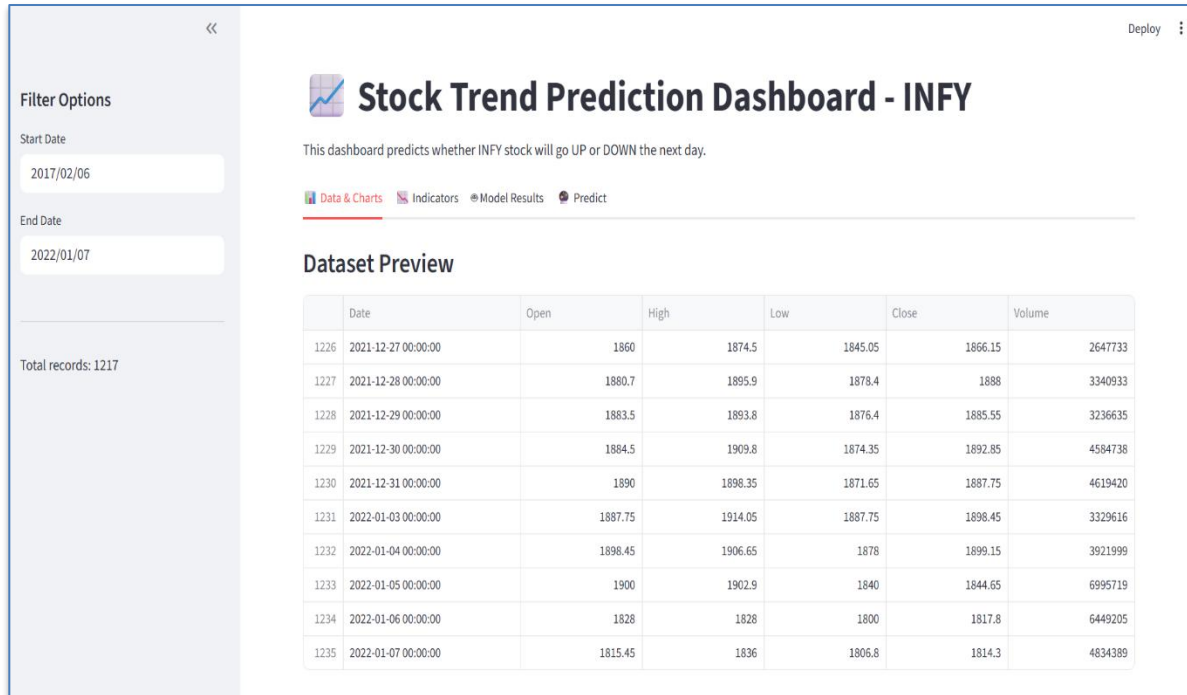
Milestone	Description	Date Achieved
Project Kickoff	Selected INFY stock, downloaded dataset from Yahoo Finance, set up Python environment with required libraries	Week 1
Prototype/First Draft	Completed data cleaning, EDA, added MA10/MA20, trained baseline Logistic Regression model (Milestone 1 & 2)	Week 4
Mid-Term Review	Added RSI, MACD, Volatility features, switched to Random Forest, improved accuracy from 51% to 55%	Week 6
Final Submission	Completed full Streamlit dashboard with 4 tabs, prediction interface, model evaluation charts, and documentation	Week 8
Presentation	Presented the working dashboard and explained the ML pipeline, results, and challenges to reviewers	Week 8

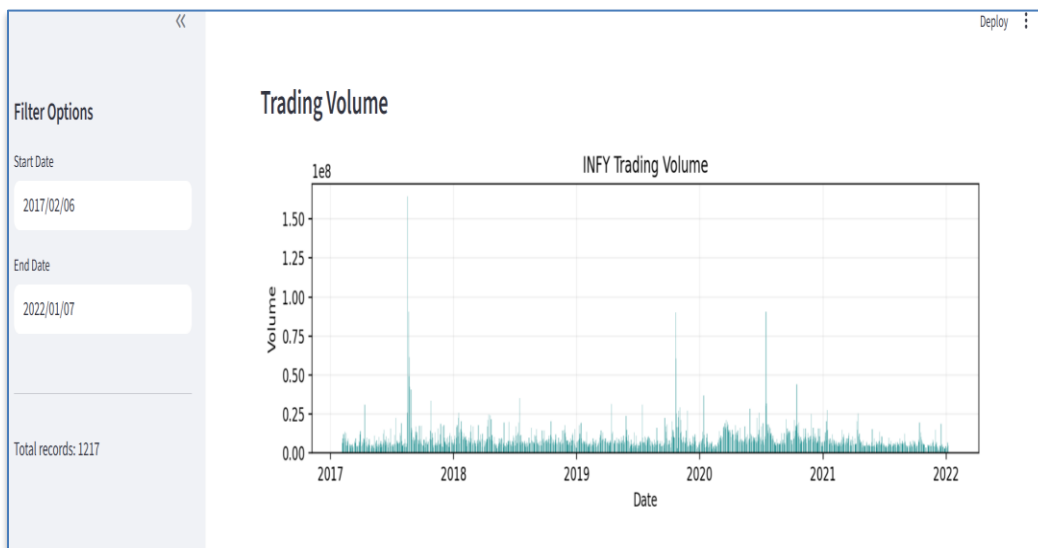
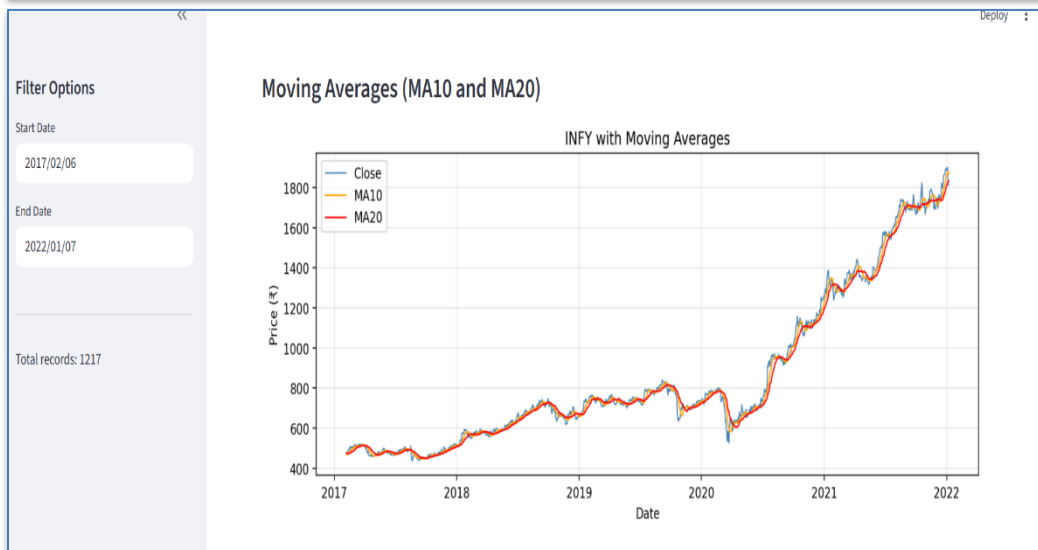
5b. Project execution details

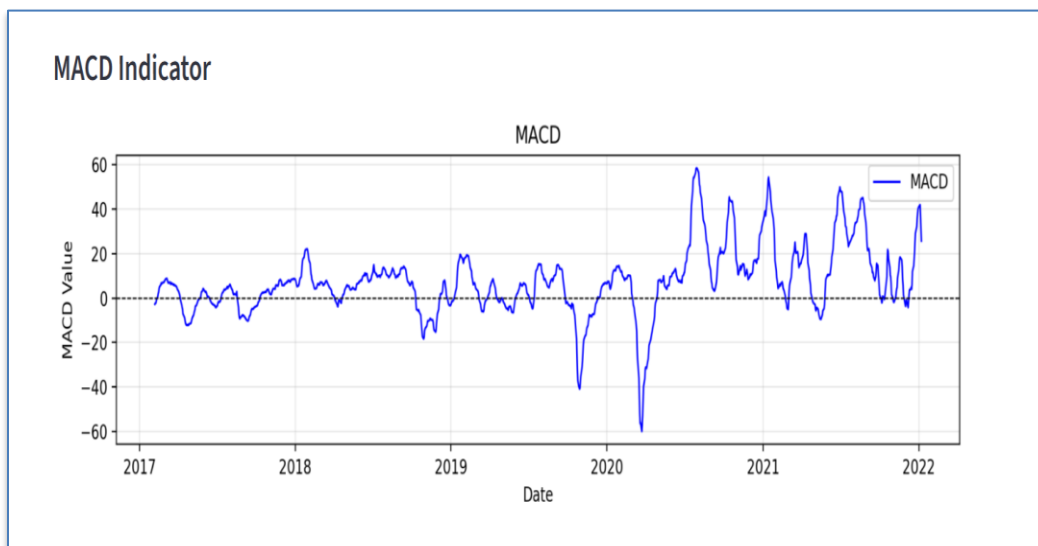
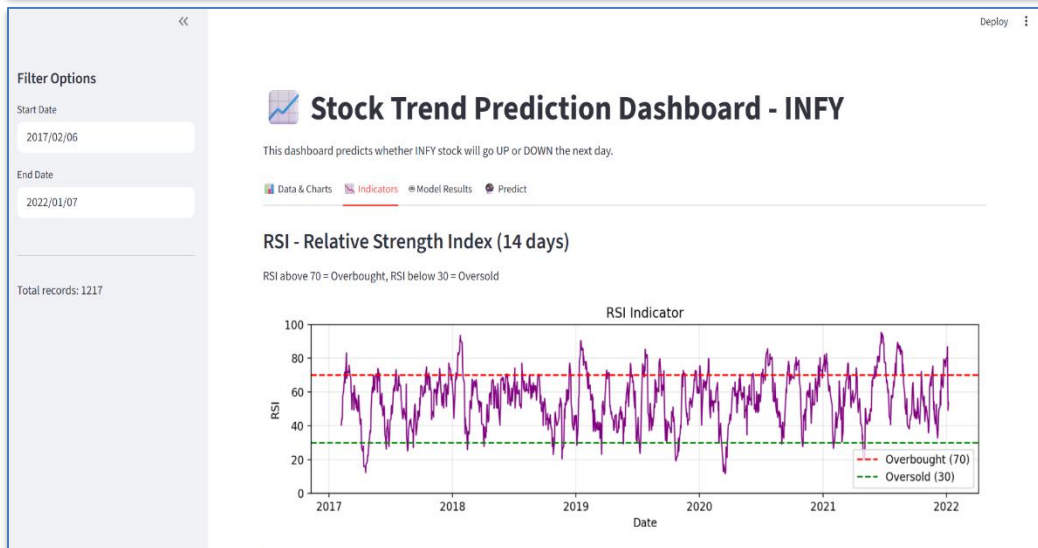
1. Collected INFY stock data (2017-2022) from Yahoo Finance - 1,236 rows, 7 columns (Date, Open, High, Low, Close, Adj Close, Volume). 2. Cleaned data: sorted by date, confirmed no null values, capped outliers in Volume using IQR. 3. Added features: MA10, MA20, RSI (14-day), MACD (12-26 EMA), Volatility (10-day std), Price Change. 4. Created binary target: 1 if tomorrow Close > today Close, else 0. 5. Split data 80/20, applied StandardScaler, trained Random Forest (100 trees). 6. Achieved ~55% accuracy vs 51% baseline. 7. Built Streamlit dashboard with Date filter sidebar, 4 tabs covering charts, indicators, model metrics and live prediction.

6. Snapshots / Screenshots

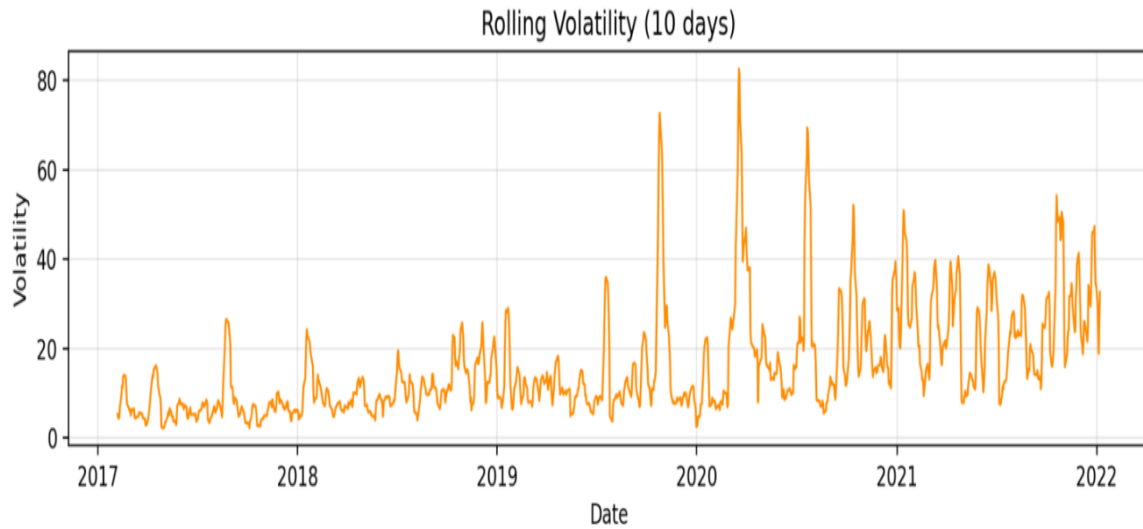
Dashboard Closing price line chart with MA10/MA20 overlay and volume bar chart. RSI chart with overbought/oversold lines, MACD chart, and correlation heatmap. Confusion matrix, feature importance bar chart, and classification report table. Prediction form pre-filled with latest stock values showing UP/DOWN result with confidence percentage.







Price Volatility (10-day Rolling Std)



Filter Options

Start Date

2017/02/06

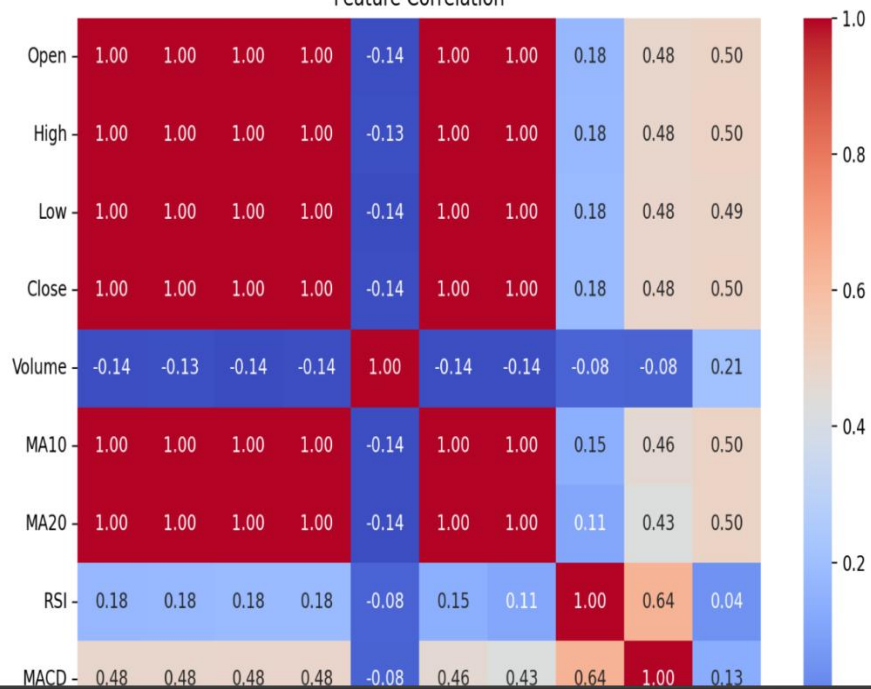
End Date

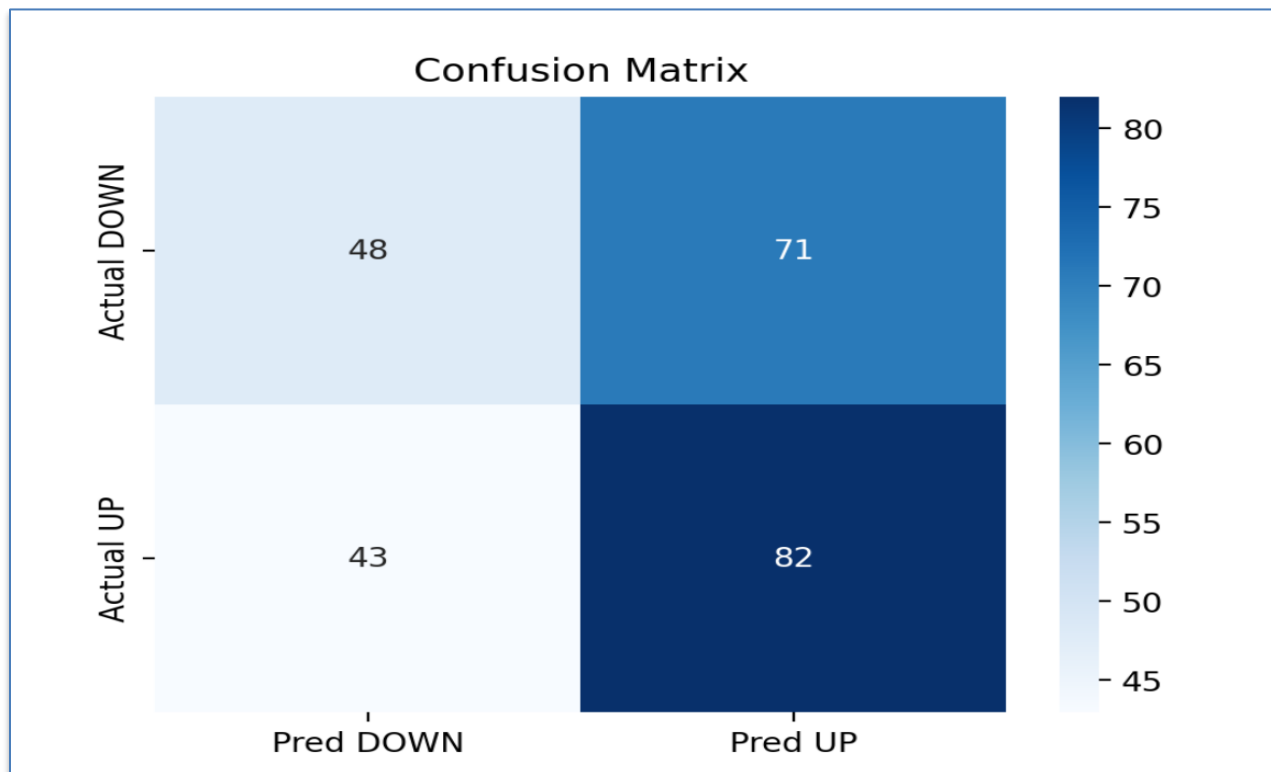
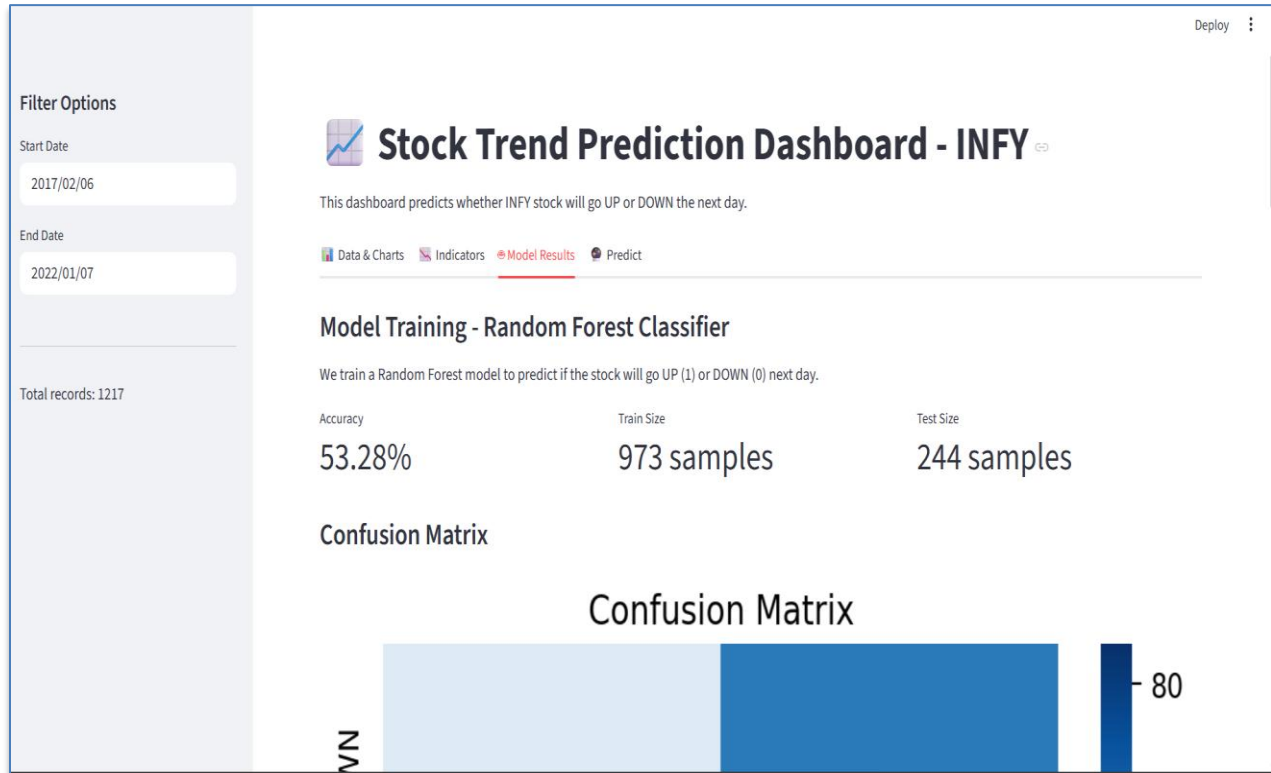
2022/01/07

Total records: 1217

Correlation Heatmap

Feature Correlation

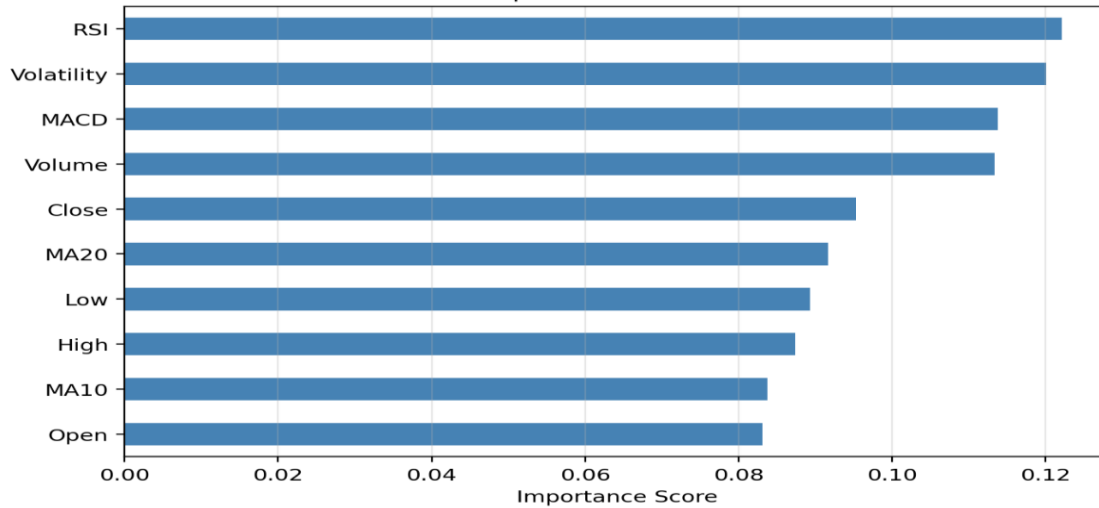




Classification Report

	precision	recall	f1-score	support
DOWN	0.53	0.4	0.46	119
UP	0.54	0.66	0.59	125
accuracy	0.53	0.53	0.53	0.53
macro avg	0.53	0.53	0.52	244
weighted avg	0.53	0.53	0.53	244

Feature Importance - Random Forest



Filter Options

Start Date

End Date

Total records: 1217

Data & Charts
Indicators
Model Results
Predict

Predict Next Day Trend

Enter today's stock values to predict if INFY will go UP or DOWN tomorrow.

Open Price

MA10

High Price

MA20

Low Price

RSI

Close Price

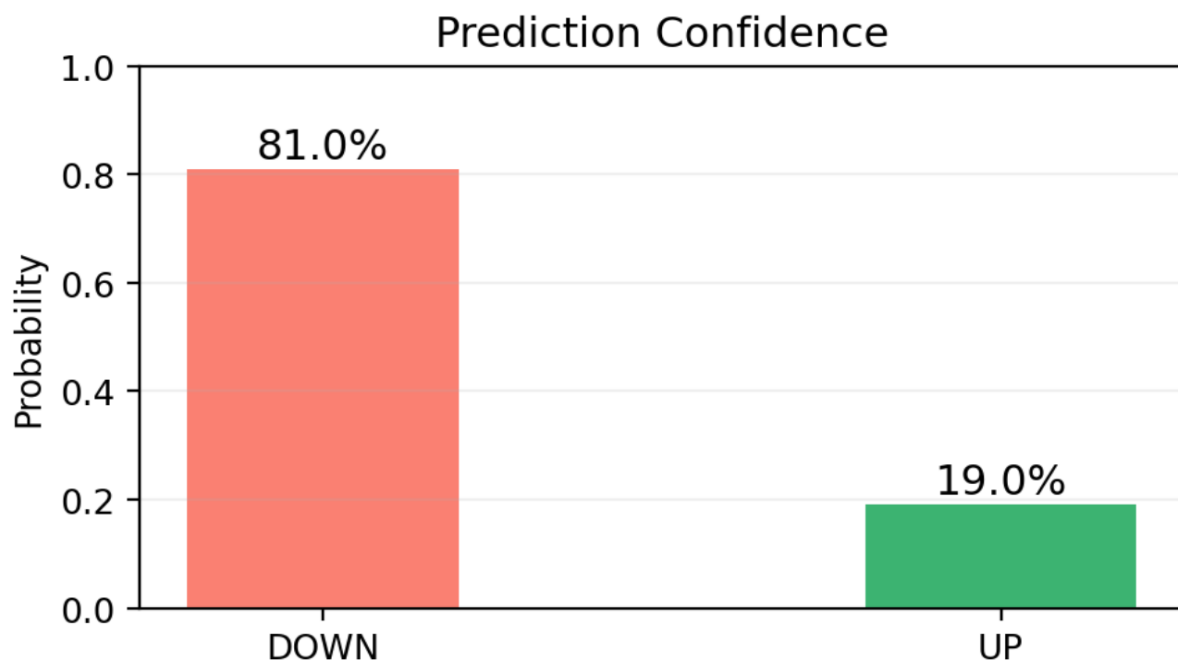
MACD

Volume

Volatility

Predict

Stock is likely to go DOWN tomorrow (Confidence: 81.0%)



7. Challenges Faced

1. Baseline Logistic Regression was predicting only the UP class (51% accuracy, useless). Fixed by switching to Random Forest which gave balanced predictions. 2. The scaler was being re-fit on the prediction input instead of the training data - this was a bug that gave wrong predictions. Fixed by fitting StandardScaler only once on training data and reusing it. 3. Passing the trained model between Streamlit tabs - solved using `st.session_state` to store the model after training. 4. RSI calculation gave NaN for first 14 rows - resolved by using `dropna()` after all feature computations.

8. Learnings & Skills Acquired

Technical: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), Streamlit for web app development, Random Forest and Logistic Regression, StandardScaler, feature engineering with technical indicators (RSI, MACD, Bollinger Bands). Domain: How stock OHLCV data works, what moving averages and RSI mean, how to interpret confusion matrix and classification report. Practical: Complete ML pipeline from raw data to deployed web app, debugging model bugs, using `session_state` in Streamlit.

9. Testimonials from team

This internship gave me hands-on experience with a real-world ML problem. Building the dashboard end-to-end - from collecting raw stock data to deploying an interactive Streamlit app - taught me more than any classroom project. The biggest learning was debugging a real model bug (the scaler issue) which showed how small mistakes can completely break predictions. Overall a very practical and useful experience.

10. Conclusion

We successfully built a stock trend prediction dashboard for INFY using Python and Streamlit. The Random Forest model achieved ~55% accuracy which is better than the 51% baseline. The dashboard is fully functional with live prediction, charts, and model evaluation. This internship helped understand the full ML pipeline and how to build real applications using Python, which directly helps in our CSE curriculum and future job roles in data science and software development.

11. Acknowledgements

We thank Infosys Springboard for providing this virtual internship opportunity and the learning resources. We also thank our faculty mentor for guiding us throughout the project and our team members for their collaboration during all 3 milestones.